

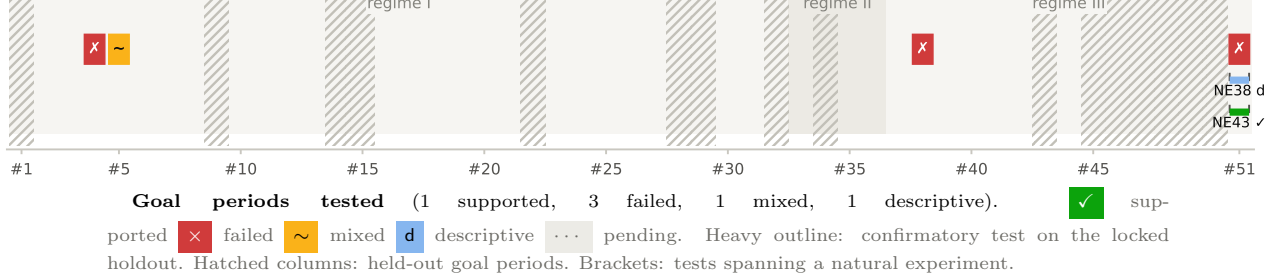
H59: One lever model for all operator inputs

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 0.5/5 Usefulness 2/5

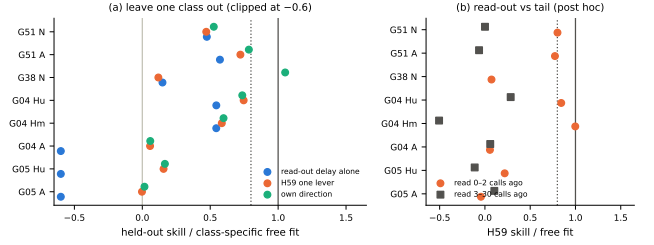
Bottom line. One triple per input class fails to predict a held-out class; classes share only the read-out-call response and differ in their tails.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The claim: one triple describes every operator input, namely field strength h , catalytic strength κ and read-out delay d . A model fitted on three input classes should then predict the fourth. Each model call is idle, work or talk. Every nudge, human message (broadcast or named) and @-mention is timed at the call that first reads it (DQ1 ledger). Per period, one linear response with a shared kernel and field direction fits all classes. Each class is held out in turn (5 day folds). Replication: G51, G38, G04; natives: NE43, #5 (G05), NE38. **The claim fails.** Held-out-class skill is 0.47 (nudges) and 0.72 (mentions) of a class-specific fit in G51, 0.06–0.75 in G04 and 0.12 for G38 nudges. At the read-out itself (0–2 calls), the shared shape keeps 77–100% of the skill in G51 and for G04 humans. The classes differ in their tails. The holdout is not run.



Observables. (a) Held-out skill of each rival relative to the class-specific free fit, per period and class (informative cells only). H59 (orange) is below the 0.8 line everywhere. The read-out-delay rival (blue) is far below for message classes. (b) Post hoc split. At the read-out (rows read 0–2 calls ago), H59 transfers for G51 nudges and mentions and for G04 human messages. In the tail (3–30 calls) it does not.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Treat the immediate pull of an operator input as portable and its aftermath as class-specific. At the receiving call, every class pulls the agent toward talk, with a little extra traffic. A nudge and an @-mention look alike there ($I \rightarrow T + 2.9$ for both in G51). One number per class plus the delay to the next model call therefore describes most of the immediate effect. The aftermath differs. Nudged agents stay idle-prone for 30 calls; this looks like the nudger selecting stuck agents, not a lever effect. Regime-I mentions speed up the work $\leftrightarrow$ wait edge. Several messages read in one call count as one read (G05). The class levers do not change when the nudger stops (NE43).

PHYSICAL MODEL

Kinetic response (model 02) with a field-catalyst split (H39), and kicks with a call-clock response kernel (model 09). Each transition's log-odds against staying has a base-line (agent, day, aging, Markov-2) plus a kick term. H59 gives each class a uniform catalyst κ_c and a field h_c along one shared direction θ . One shared kernel K_ℓ runs over lags since the receiving call. Rivals: class-specific free responses, a class-specific direction, the read-out delay alone, the inert model.

$$\eta_{ij} = \alpha_{ij} \cdot x + \sum_{c,\ell} D_{c\ell} \beta_{ij}(c, \ell)$$

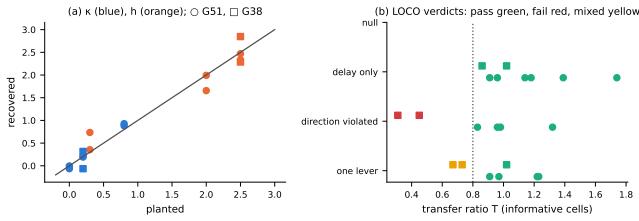
$$\text{H59: } \beta_{ij}(c, \ell) = K_\ell \left[\kappa_c + \frac{h_c}{2} (u_j - u_i) \right], \quad u = (0, \cos \theta, \sin \theta)$$

$$T_c = S_{\text{H59}} / S_{\text{free}}, \quad S = \text{held-out log-lik gain over inert}$$

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RESULTS VS PREDICTIONS

Prediction	Observed	
P1 LOCO $T \geq 0.8$ (G51)	nudges 0.47, mentions 0.72	×
P2 delay alone loses	messages yes; nudges tie	~
P3 kernel at read-out	K_1 0.44 (G51), 0.65 (G38)	~
P4 no inbox decay	nudges, mentions yes; broadcast humans 0.57	~
P5 lever-table signs	all but broadcast-human field size	✓
P6 G38, G04	both fail	×
P7 synthetic	recovery yes; direction blind at idle	~
NE43 invariance	Δ CIs $\ni 0$; transfer 1.01	✓
G05 dose saturates	≥ 3 vs 1 msg: -0.02 ; LOCO fails	~



Second figure. Synthetic validation with the real exposure design. (a) The fit recovers κ and h at G51 and G38 counts. (b) LOCO verdicts by planted world: a one-lever world passes at G51. The test catches a direction violation at G38 but not at G51. In G51 nudges reach almost only idle agents, where two parameters fit any direction.

GOAL PERIODS IN DETAIL

G04 × failed
G05 ~ mixed
G38 × failed
G51 × failed
NE38 d descriptive
NE43 ✓ supported

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`witeup/paper.tex`).

CAVEATS

The read-out/tail split is post hoc; it is frozen for the holdout as C1. The state space is behavioral only and holds no content fields. The baseline is fitted once and held fixed in the bootstraps. Named-human (92 reads) and G38 broadcast-human cells are uninformative. Kickoffs are not testable at call level. The nudge tail may be the nudger's targeting, which the past-only baseline cannot see.

ROUND 2

Run `confirm.py` on the #51 tail. Test whether the nudge tail is selection (trigger state, first fire against re-fire). Try a two-component lever: an edge catalyst plus a second direction. Add content drift on the same clock. Pool kickoffs across regime-III periods.

- **Essence.** a compact, portable description of what an operator input does. Round 1 says the portable part is the read-out response, and what differs between classes is their tails and, in some regimes, which edge they act on.
- **H59-R1** Run `confirm.py` on the #51 tail after commit and disclosure.
- **H59-R2** Separate tail from selection.
- **H59-R3** Two-component lever.
- **H59-R4** Content layer.
- **H59-R5** Kickoffs across regime-III periods with partial pooling (exception d), as the fifth class.

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